

Activity – 9

1. C program to input 2 numbers , swap them using third variable and swapping them without using third variable.

```
#include <stdio.h>
int main(){
    int num1, num2, temp, choice;
    printf("Enter two numbers.\n");
    scanf("%d %d", &num1, &num2);

    do {
        printf("\n —— Swap Menu —— \n");
        printf("\nSwap using third variable\n");
        printf("\nSwap using without any third variable\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice){
            case 1:
                printf("\n——By using third variable——\n");
                temp = num1;
                num1 = num2;
                num2 = temp;
                printf("After swapping num1 = %d, num2 = %d\n", num1, num2);
                break;
            case 2:
                printf("\n——By without third variable——\n");
                num1 = num1 + num2;
                num2 = num1 - num2;
                num1 = num1 - num2;
                printf("After swapping num1 = %d, num2 = %d\n", num1, num2);
                break;
            case 3:
                printf("\nExiting program by pressing 3\n");
                break;
            default:
                printf("invalid choice please try again\n");
        }
    }while(choice != 3);
    return 0;
}
```

Compilation and execution

```
(kali㉿kali)-[~]
$ vi newd1.c

(kali㉿kali)-[~]
$ gcc newd1.c -o newd1

(kali㉿kali)-[~]
$ ./newd1
Enter two numbers.
18 20

  _____ Swap Menu _____
Swap using third variable
Swap using without any third variable
Enter your choice: 1

  _____By using third variable_____
After swapping num1 = 20, num2 = 18

  _____ Swap Menu _____
Swap using third variable
Swap using without any third variable
Enter your choice: 2

  _____By without third variable_____
After swapping num1 = 18, num2 = 20

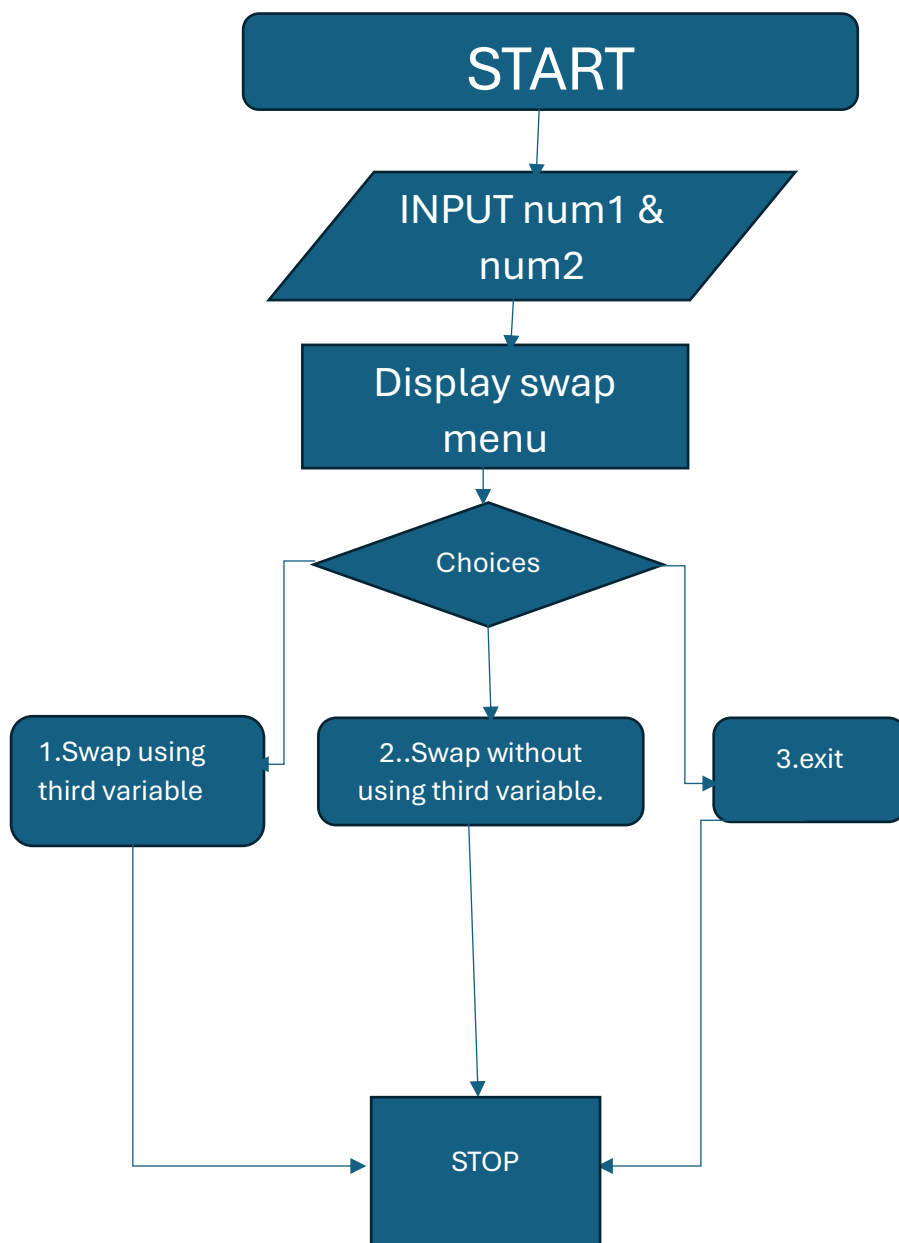
  _____ Swap Menu _____
Swap using third variable
Swap using without any third variable
Enter your choice: 3
Exiting program by pressing 3
```

Algorithm

1. Start
2. Take input of num1 and num2 from user.
3. Make 3 choices – by swapping , by non swapping and exiting.

4. Use do while loop and switch cases (1, 2, 3)
5. Print the values according to choices.
6. STOP

FLOWCHART



PSEUDOCODE

1. START
2. DECLARE INT NUM1, NUM2,
TEMP, choice
3. Input num1, num2
4. Input choice
5. If choice == 1
Do by swapping
If choice == 2
Do without swapping
If choice == 3
Exit
6. STOP

2. C program to find sum of first n natural numbers and square of n natural numbers.

```
#include <stdio.h>
int main(){
    int n, sum1, sum2;
    printf("Enter the number n;");
    scanf("%d", &n);

    sum2 = n*(n+1)*(2*n+1)/6;
    sum1 = n * (n + 1) / 2;
    printf("Sum of first n natural numbers:%d", sum1);
    printf("Sum of squares of first n natural numbers:%d", sum2);
    return 0;
}
```

Compilation and execution

```
(kali㉿kali)-[~]
$ vi newd2.c

(kali㉿kali)-[~]
$ gcc newd2.c -o newd2

(kali㉿kali)-[~]
$ ./newd2
Enter the number n;2
Sum of first n natural numbers:3Sum of squares of first n natural numbers:5
```

Algorithm

1.START

2.input of n

3.sum1, sum2 for sum and square sum

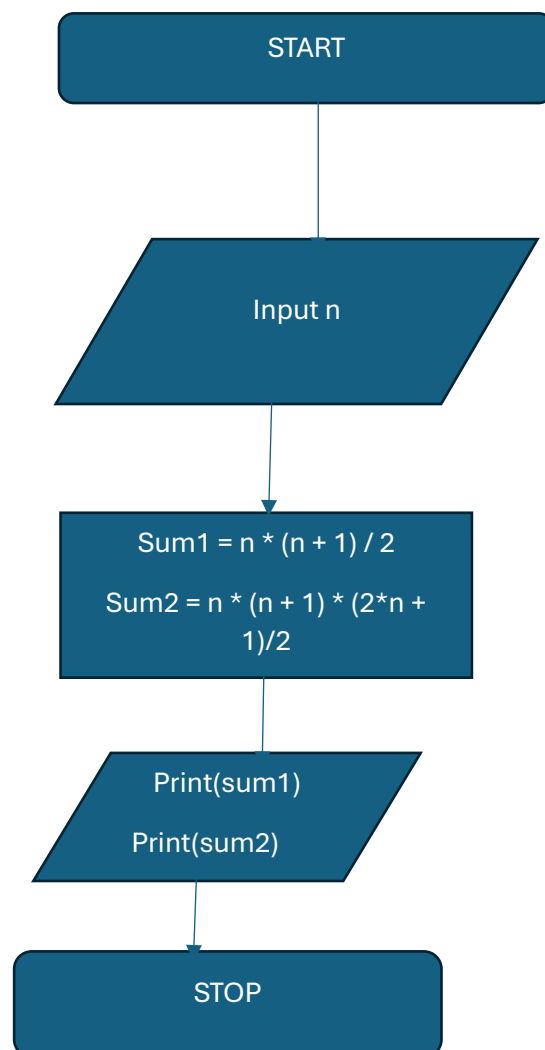
4. $\text{sum1} = n * (n+1) / 2$

5. $\text{sum2} = n * (n+1) * (2n+1) / 2$

6.print sum1, sum2

7.STOP

Flowchart



Pseudocode

1. START
2. Declare int n, sum1, sum2;
3. Input n
4. $\text{Sum1} = n * (n + 1) / 2$
5. $\text{Sum2} = n * (n + 1) * (2 * n + 1) / 6$
6. Print(Sum1)
7. Print(Sum2)
8. STOP

3. C program to calculate compound interest, compound amount, total payable amount after given time, rate, principal.

```
#include <stdio.h>
#include <math.h>
int main()
{
    double principal, rate, time, amount, compound;

    printf("Enter principal amount:");
    scanf("%lf", &principal);

    printf("Enter annual interest rate:");
    scanf("%lf", &rate);

    printf("Enter time:");
    scanf("%lf", &time);

    amount = principal * pow((1 + rate / 100), time);

    compound = amount - principal;

    printf("\n--- Results ---\n");
    printf("total payable amount (A) = %.21f\n", amount);
    printf("Compound Interest (CI) = %.21f\n", compound);

    return 0;
}
```

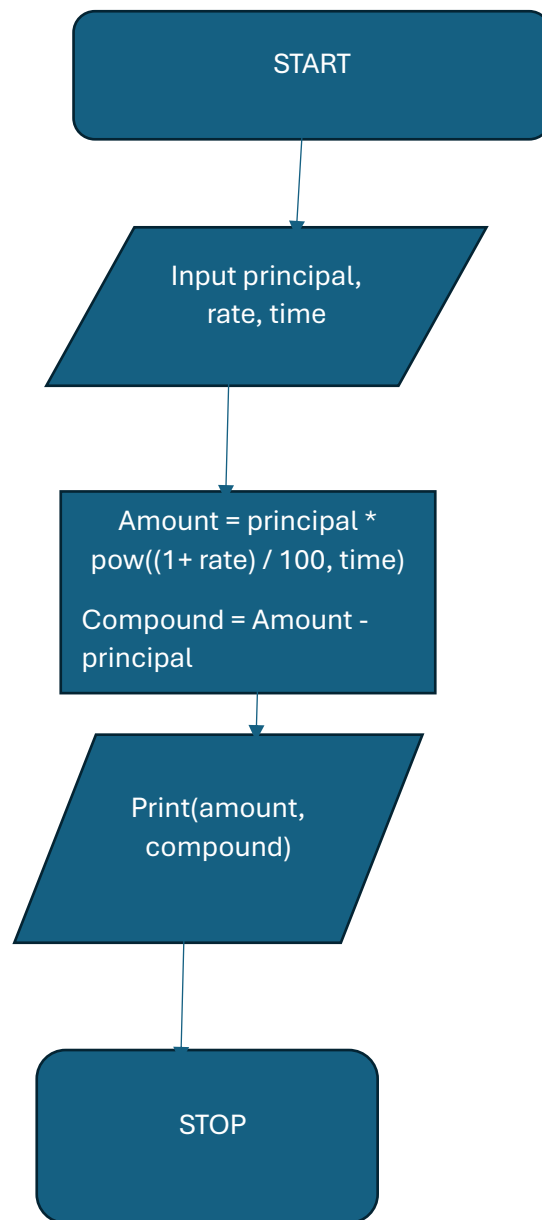
Compilation and execution

```
(kali㉿kali)-[~]  
$ vi newd3.c  
  
(kali㉿kali)-[~]  
$ gcc newd3.c -o newd3 -lm  
  
(kali㉿kali)-[~]  
$ ./newd3  
Enter principal amount:1200  
Enter annual interest rate:2  
Enter time:1  
  
— Results —  
total payable amount (A) = 1224.000000000000000000000000  
Compound Interest (CI) = 24.000000000000000000000000
```

Algorithm

1. START
2. Input principal, rate, time
3. $\text{Amount} = \text{principal} * \text{pow}((1 + r/100), \text{time})$
4. $\text{Compound} = \text{Amount} - \text{principal}$
5. Print amount, compound interest
6. STOP

Flowchart



Pseudocode

1. START
2. Declare int principal, rate, time, compound, amount

3. $\text{Amount} = \text{principal} * \text{pow}((1 + \text{rate}/100, \text{time}))$
4. $\text{Compound} = \text{Amount} - \text{principal}$
5. $\text{Print}(\text{amount}, \text{Compound})$
6. STOP

